01_CATALOG_DOMAIN_CONCEPTUAL_ENTITY_PASS.md

01_CATALOG_DOMAIN.md

Purpose

The Catalog Domain represents conceptual origami knowledge and publication structure.

It provides the canonical identity and organization of works, editions, titles, revisions, and explicit relationships among works.

The Catalog Domain is the identity foundation used by Attribution, Rights, Instruction, Artifact, Provenance, Taxonomy, External Reference, and Community domains.

Scope

The Catalog Domain answers:

- What conceptual origami design is being described?
- Which distinct published, translated, reconstructed, adapted, or revised expressions exist for that design?
- Which titles are associated with the design?
- How is the design related to other designs?
- Which significant revisions belong to a particular edition?

The Catalog Domain does not own:

- Parties or credits.
- Licenses or rights statements.
- Instruction sets or steps.
- Digital files or artifact versions.
- FOLD documents or frames.

- Import, validation, transformation, or value-provenance records.
- Techniques, bases, categories, or tags.
- Community accounts or participation records.

Those concepts are owned by their respective domains and may reference Catalog entities.

Domain Ownership

The Catalog Domain owns:

- `OrigamiWork`
- `OrigamiEdition`
- `WorkTitle`
- `WorkRelationship`
- `Revision`

Core Conceptual Decisions

Work Is the Stable Conceptual Identity

`OrigamiWork` represents the identity of a conceptual origami design.

A work is independent of:

- A particular title.
- A particular language.
- A publication or release.
- A set of instructions.

- A diagram, photograph, video, PDF, FOLD file, or other artifact.
- A particular folded specimen created by a community member.

The work may be recognized through its intended design identity, but the Catalog Domain does not define that identity solely as a folding sequence, final physical specimen, geometric state, or file.

Edition Is a Distinct Expression of One Work

`OrigamiEdition` represents one distinct published, translated, reconstructed, adapted, or otherwise curated expression of exactly one `OrigamiWork`.

An edition may organize or contextualize instructions, artifacts, credits, rights, and external references, but those records remain owned by their respective domains.

Title Is Separate from Work Identity

`WorkTitle` owns title values. A title is not the identity of the work.

This allows alternate, historical, translated, and multilingual titles to coexist without creating duplicate works.

Relationships Are Explicit Claims

`WorkRelationship` records an explicit relationship claim between two works. Relationships are not inferred merely from similar names, geometry, subjects, bases, or techniques.

Revision Is Edition-Level Catalog History

`Revision` records a named or significant revision of an edition.

Revision does not replace artifact versioning, transformation provenance, audit history, or value-level provenance.

Entity: OrigamiWork

Definition

A stable Catalog identity representing one conceptual origami design.

Identity Boundary

`OrigamiWork` answers:

Code — Plain Text

```
Which conceptual design is this?

```

It does not answer:

Code — Plain Text

```
Which publication is this?
Which file is this?
Which instructional workflow is this?
Which folded specimen is this?
Which geometric frame is this?

```

Conceptual Attributes

- `id`
- `slug`
- `summary`
- `status`

- `notes`
- `created_at`
- `updated_at`

The preferred display title is resolved through `WorkTitle`; it is not a second authoritative title field on `OrigamiWork`.

Relationships

Code Plain Text

```
OrigamiWork
    1 ─── 0..* OrigamiEdition
    1 ─── 0..* WorkTitle
    1 ─── 0..* WorkRelationship (outgoing)
    1 ─── 0..* WorkRelationship (incoming)

```

Cross-domain relationships may include credits, classifications, external references, community activity, and other records owned outside Catalog.

Integrity Rules

- A published work must have at least one `WorkTitle`.
- A draft or preservation-first work may temporarily have incomplete metadata.
- Missing attribution, license, classification, geometry, or instructions must not invalidate the work.
- Imported title or author text must not silently overwrite curated Catalog or Attribution records.
- Similarity alone must not automatically merge two work identities.
- A merge or split of work identities requires an explicit curated process with preserved provenance.

Cross-Work Versus Non-Work

Same Work Versus New Work

The following normally remain the same work:

- A translated title.
- A new diagram set.
- A new instructional language.
- A new video or photograph.
- A corrected file.
- A new publication of the same conceptual design.

The following may justify a new work when the design identity is intentionally changed:

- A substantial variation.
- A simplification that changes the design identity.
- An expansion or remix recognized as a distinct design.
- A derivative design with independent identity.

When uncertain, preserve separate records and record a `WorkRelationship` rather than forcing an irreversible merge.

Entity: OrigamiEdition

Definition

A distinct published, translated, reconstructed, adapted, revised, or otherwise curated expression of exactly one `OrigamiWork`.

Identity Boundary

`OrigamiEdition` answers:

Code Plain Text

```
Which expression or release of the work is this?

```

An edition may correspond to:

- An original publication.
- A translation.
- A reconstructed presentation.
- A revised instructional release.
- An interactive or mixed-media presentation.
- A community-curated edition.

A container such as a book, magazine, or website is not automatically an edition by itself. The edition is the expression of the work represented or published through that container.

Conceptual Attributes

- `id`
- `work_id`
- `title`
- `edition_type`
- `summary`
- `language`
- `publication_date`
- `status`
- `created_at`
- `updated_at`

The edition title may identify a release or presentation. It does not replace the work's `WorkTitle` records.

Relationships

Plain Text

```
OrigamiWork
    1 ——— 0..* OrigamiEdition

OrigamiEdition
    * ——— 1 OrigamiWork
    1 ——— 0..* Revision

```

Cross-domain relationships may include credits, licenses, instruction sets, digital artifacts, external references, completed folds, and provenance records.

Integrity Rules

- Every edition belongs to exactly one work.
- An edition must not belong to multiple works.
- A work may exist without an edition during preservation or curation.
- Replacing, correcting, or deriving a file does not by itself create a new edition.
- A language change, substantive instructional reconstruction, or intentionally distinct publication expression may create a new edition.
- Imported file metadata may suggest an edition but does not automatically establish one as verified Catalog knowledge.

New Edition Versus New Artifact Version

Create or consider a new edition when the expression itself changes, for example:

- Translation into another language.
- Substantive reconstruction of instruA plugin for IntelliJ IDEA, PyCharm, WebStorm, actions.

- Deliberately distinct release or presentation.
- Material editorial revision intended as a new expression.

Create a new `ArtifactVersion`, not a new edition, when only the preserved file changes, for example:

- Corrected scan.
- Re-encoded video.
- Re-exported PDF.
- New checksum-equivalent preservation copy.
- Generated thumbnail or preview.

Artifact-version semantics are owned by the Artifact Domain.

Entity: WorkTitle

Definition

A title associated with one `OrigamiWork`.

Purpose

`WorkTitle` preserves:

- Primary titles.
- Alternative titles.
- Historical titles.
- Translated titles.
- Aliases.
- Multiple preferred display titles for different languages or contexts.

Conceptual Attributes

- `id`
- `work_id`
- `title`
- `language`
- `title_type`
- `preferred`
- `created_at`
- `updated_at`

Suggested Title Types

- `Primary`
- `Alternative`
- `Historical`
- `Translated`
- `Alias`

Relationship

Code — Plain Text

```
OrigamiWork
    1 ─── 0..* WorkTitle

WorkTitle
    * ─── 1 OrigamiWork

```

Integrity Rules

- Every title belongs to exactly one work.

- A work may have multiple titles.
- At most one title should be preferred for the same work and language within the same preference context.
- Historical and alternate titles should be preserved rather than overwritten.
- Matching title text does not prove that two records represent the same work.
- Imported title text remains a source claim until curated into a `WorkTitle`.

Entity: WorkRelationship

Definition

An explicit relationship claim connecting a source work to a target work.

Conceptual Attributes

- `id`
- `source_work_id`
- `target_work_id`
- `relationship_type`
- `notes`
- `status`
- `created_at`
- `updated_at`

Provenance for the relationship claim is owned by the Provenance Domain and may be attached through cross-domain records.

Initial Controlled Relationship Types

- `Variation Of`
- `Derived From`
- `Remix Of`
- `Simplification Of`
- `Expansion Of`
- `Related To`

The vocabulary may be extended through controlled records rather than arbitrary free text.

Directionality

Directional relationships are interpreted as:

Code Plain Text

```
source_work_id → target_work_id

```

Examples:

Code Plain Text

```
source Variation Of target
source Derived From target
source Simplification Of target
source Expansion Of target
source Remix Of target

```

`Related To` is symmetric in meaning. Its storage strategy is deferred to the relational schema.

Relationships

Code Plain Text

```
OrigamiWork
    1 ——— 0..* WorkRelationship (as source)
    1 ——— 0..* WorkRelationship (as target)

WorkRelationship
    * ——— 1 OrigamiWork (source)
    * ——— 1 OrigamiWork (target)

```

Integrity Rules

- Source and target must both identify existing works.
- Source and target must not be the same work.
- Relationship type must be controlled.
- A relationship may be unverified, disputed, curated, or verified through status and provenance mechanisms.
- Conflicting relationship claims may coexist when their provenance and status remain distinguishable.
- A relationship must not be inferred solely from matching titles, shared technique, shared base, similar geometry, or imported metadata.
- Duplicate equivalent relationships should be prevented or reconciled during curation.

Entity: Revision

Definition

A preserved record of a named or significant revision to one `OrigamiEdition`.

Ownership Decision

`Revision` remains Catalog-owned because it describes the revision identity and history of an edition.

The Provenance Domain owns the evidence and process records explaining who changed what, how, why, and from which source. The Artifact Domain owns immutable file versions and derivations.

ns and derivations.Conceptual Attributes

- `id`
- `edition_id`
- `revision_number`
- `summary`
- `created_at`

Additional audit and provenance fields are defined outside Catalog.

Relationship

Code Plain Text

```
OrigamiEdition
    1 ——— 0..* Revision

Revision
    * ——— 1 OrigamiEdition

```

Integrity Rules

- Every revision belongs to exactly one edition.
- Revision numbering or labels must be unique within an edition when

supplied.

- A revision records significant edition-level change; it is not a substitute for an audit log.
- A corrected or replaced file is represented through `ArtifactVersion` unless the edition expression itself changed.
- A materially distinct expression may require a new edition rather than another revision.
- Superseded revisions may remain preserved.

Revision Versus New Edition

Use a revision when:

- The same edition identity is retained.
- Corrections or bounded editorial changes are recorded.
- The release is explicitly presented as a revision of the edition.

Use a new edition when:

- The expression is intentionally distinct.
- Translation changes the instructional language.
- Instructions are substantially reconstructed.
- The release has an independent publication identity.

The exact operational threshold is a curation policy and may evolve without changing the conceptual distinction.

Consolidated Relationships

Code — Plain Text

```
OrigamiWork
    1 ─── 0..* WorkTitle
```

```
    1 ——— 0..* OrigamiEdition
    1 ——— 0..* WorkRelationship (outgoing)
    1 ——— 0..* WorkRelationship (incoming)

OrigamiEdition
    * ——— 1 OrigamiWork
    1 ——— 0..* Revision

WorkTitle
    * ——— 1 OrigamiWork

WorkRelationship
    * ——— 1 OrigamiWork (source)
    * ——— 1 OrigamiWork (target)

Revision
    * ——— 1 OrigamiEdition

```

Cross-domain references:

Code — Plain Text

```
OrigamiWork
    ├── Credit
    ├── Category
    ├── Tag
    └── Community records

OrigamiEdition
    ├── Credit
    ├── License
    ├── InstructionSet
    ├── DigitalArtifact
    ├── ExternalReference
    ├── CompletedFold
    └── Provenance records

```

The referenced concepts remain owned by their respective domains.

Domain Boundaries

Catalog owns:

- Work identity.
- Edition identity.
- Work titles.
- Work-to-work relationship claims.
- Edition-level revision identity.

Catalog does not own:

- Creator identity or credit semantics.
- Licensing or permissions.
- Instructional sequence or pedagogy.
- Artifact bytes, versions, checksums, or derivations.
- FOLD geometry or frames.
- Provenance evidence, transformations, or validation reports.
- Taxonomy vocabulary.
- Community identity or participation.

Conceptual Integrity Summary

Code Plain Text

```
OrigamiWork ≠ OrigamiEdition ≠ DigitalArtifact

WorkTitle identifies a work by name;
it does not constitute the work's identity.

Revision belongs to OrigamiEdition;
ArtifactVersion belongs to DigitalArtifact.

WorkRelationship is explicit;
```

```
workRelationship is explicit;
similarity does not create it automatically.

```

MVP Scope

MVP Catalog entities:

- `OrigamiWork`
- `OrigamiEdition`
- `WorkTitle`
- `WorkRelationship`
- `Revision`

MVP workflow may progressively collect metadata, but every record that exists must satisfy its integrity rules.

Future Expansion

Potential future Catalog concepts:

- `Series`
- `WorkSeriesMembership`
- `PublicationContainer`
- `PublicationOccurrence`
- `EditionRelationship`

What Is an OrigamiWork?

A stable conceptual design identity, not merely a folding sequence, final

specimen, geometric state, title, instruction set, or file.

What Is an OrigamiEdition?

A distinct expression or release of exactly one work.

Can a File Create a New Edition?

Not by itself. File replacement or derivation normally creates an artifact version or derivation. A new edition requires a distinct expression or release identity.

Where Does Revision Belong?

Catalog owns edition-level revision identity. Provenance owns change evidence and process history. Artifact owns immutable file versions.

How Are Ambiguous Work Identities Handled?

Preserve uncertainty, avoid automatic merges, and use explicit relationships and provenance until curation resolves the identity question.

Remaining Open Questions

- Should editions have their own multilingual title entity in a later phase?
- Should `PublicationContainer` model books, magazines, websites, repositories, and events separately from editions?
- Which lifecycle statuses should be standardized for works, editions, relationships, and revisions?
- Should symmetric `Related To` relationships be stored once or materialized in both directions?

These questions do not block the core conceptual definitions.

- The distinction between Catalog revision, Artifact versioning, and Provenance history.
- The boundaries between Catalog and all referencing domains.